

**Report POC Deployment Image WebSocket to CCE Huawei Cloud
(Region Singapore)**

Report Date : 30-04-2025
Project Name : Deployment Image WebSocket to CCE Huawei Cloud
Project Detail : Deployment Image WebSocket to CCE Huawei Cloud

Resource

Node Pool

No	Name	Region	AZ	Flavor	vCPUs	Memory (GiB)	OS	Bandwith Type
1	Nodepool-messaging	Singapore	AZ1 -AZ5	C7n.2xlarge.2	8	16	Ubuntu 20.4	300Mbit/s Exclusive Shared
2	Nodepool-base	Singapore	AZ1- AZ5	C7n.2xlarge.2	8	16	Windows Server 2016 Datacenter 64bit	300Mbit/s Exclusive Shared

Cluster

Kategori	Parameter	Nilai
Basic Settings	Type	CCE Turbo
	Billing Mode	Yearly/Monthly 1 month Auto-renewal
	Cluster Name	Mandatory
	Cluster Version	v1.30
	Cluster Scale	Nodes: 50
	Master Nodes	3 Masters
	Master Distribution	Automatic
Network Settings	Network Model	Cloud Native Network 2.0
	VPC	Mandatory

	IPv6	Close
	Default Node Security Group	Auto generate
	DataPlane V2	Close
	Default Security Group	Auto generate
	Service CIDR Block	10.247.0.0/16
	Request Forwarding	iptables
Advanced Settings	Overload Control	Enable
	Time Zone	(UTC+08:00) Asia/Shanghai
Add-ons	CCE Container Network	Default Installation

Ingress/LoadBalancer

Bagian	Field	Nilai / Pilihan
General	Name	ingress-messaging
	Namespace	gandalfus-messaging
Load Balancer	Load Balancer Type	Shared
	Auto Create	✓ (enabled)
	Instance Name	elb-messaging-kubernetes
	Guaranteed Performance	✗ (not selected)
	EIP	Auto create
	Line	Dynamic BGP
	Billing Mode	Traffic (for light/bursting traffic)
	Bandwidth Options	1, 5, 10, 50, 100, 300 Mbps (custom selection area)

	Selected Bandwidth	300 Mbps
	Resource Tag	- Tag Management Services (TMS)- View TMS Tag List- Add tag (optional)
Listener	Frontend Protocol	HTTP (selected), HTTPS (not selected)
	External Port	Auto create (with note: route maintains external port config of the created listener)
	Access Control	Inherit ELB Configurations
	Advanced Options	Add advanced options (optional)
Footer	Notes	I have read Notes on Using Load Balancers
	Button	OK

Work Detail

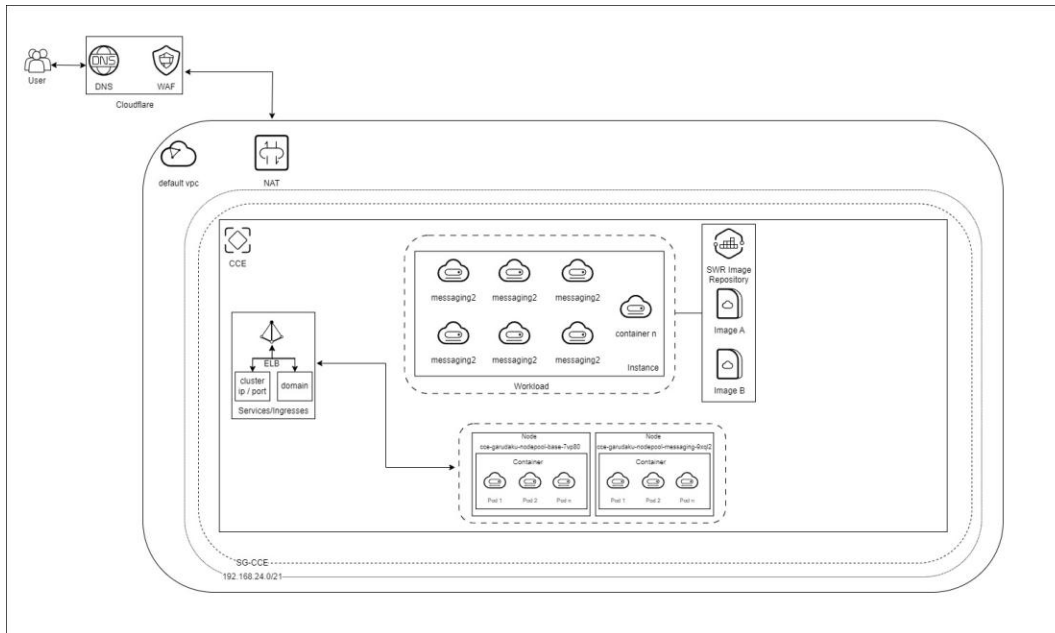
Planned start datetime	25-02-2025 08.30
Planned finish datetime	14-03-2025 23:00
Actual start datetime	25-02-2025 08:30
Actual finish datetime	14-03-2025 23:59
Delay Reason	.
Resource Used	CCE, ELB, EIP, ECS, SWR
Client Confirmation Time	14-03-2025
Release Time	

Client Detail (PIC)

Name	Phone	Organization
Mr Subangkit	+62 811-1353-979	Garudaku

Dokumentasi

Topologi



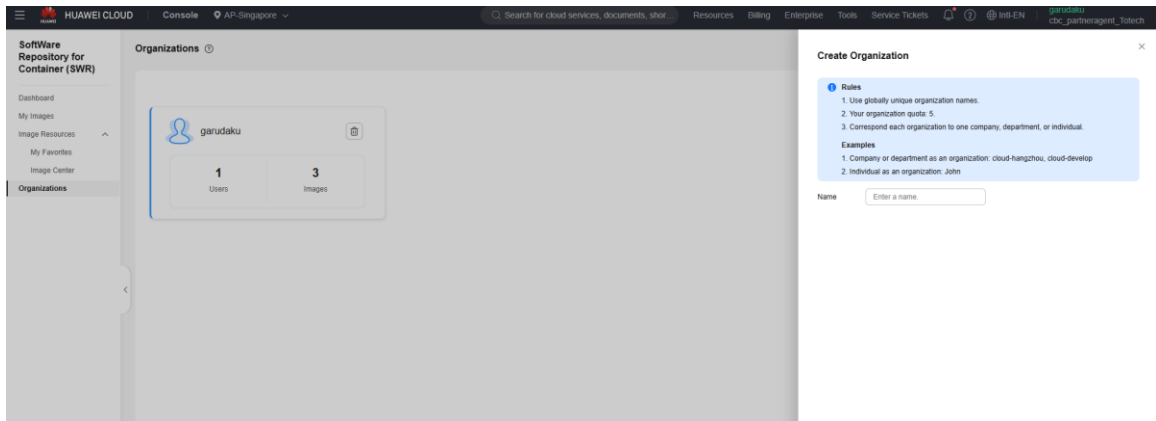
1. Prasyarat

- Akun Huawei Cloud aktif
- Image WebSocket di lokal (Docker)
- Akses ke Huawei Cloud SWR dan CCE
- Huawei Cloud CLI atau akses ke Console

2. Push Image WebSocket ke SWR Huawei Cloud

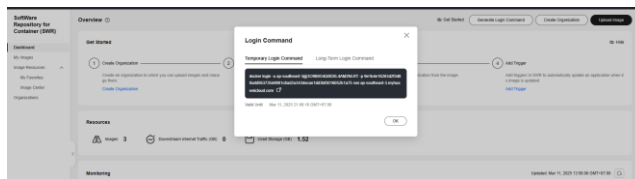
Langkah 1: Login ke SWR Huawei Cloud dan Buat Organization

1. Login ke Huawei Cloud Console melalui <https://console.huaweicloud.com>.
2. Navigasi ke menu: SoftWare Repository for Container (SWR)> Organization > Create Organization.

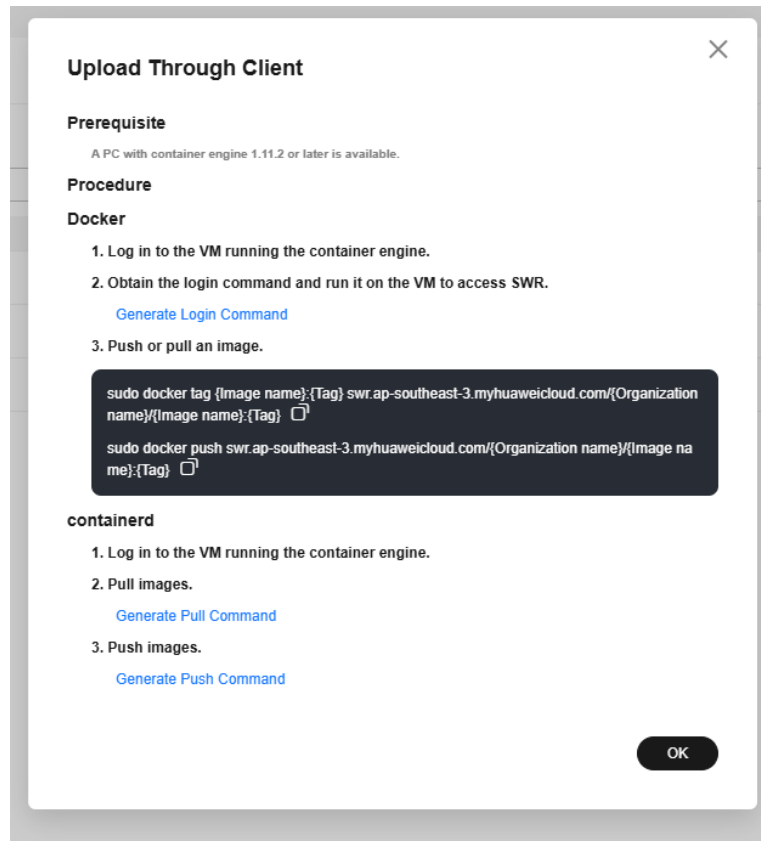


docker login swr.ap-southeast-3.myhuaweicloud.com

- Gantilah <region> dengan region tempat SWR kamu berada, misalnya: cn-north-4, ap-southeast-3, dll.
- Username dan password-nya bisa ambil dari **akses SWR kamu di Huawei Cloud** (biasanya sama dengan AK/SK jika menggunakan mode IAM).
- Bisa menggunakan Temp Login (Klik Generate Login Command)



Langkah 2: Tag Image Lokal ke SWR Format



Misalnya image lokal Anda bernama websocket-server:latest

```
docker tag websocket-server:latest swr.ap-southeast-3.myhuaweicloud.com/<namespace>/<repository>:latest
```

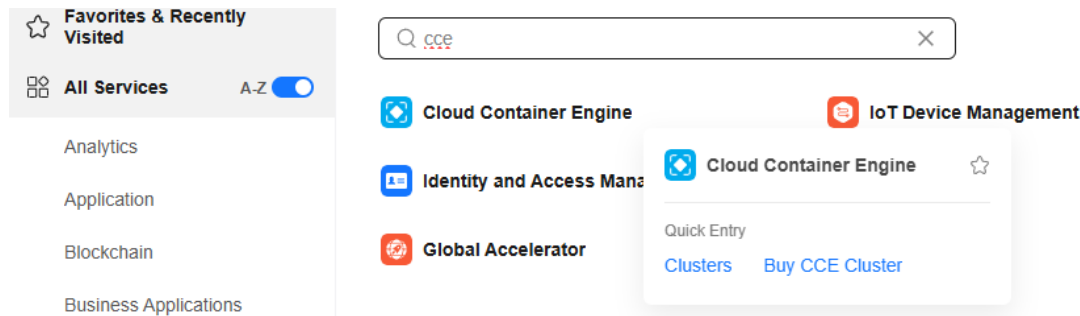
Contoh:

```
docker tag websocket-server:latest swr.ap-southeast-3.myhuaweicloud.com/aryadev/websocket-server:latest
```

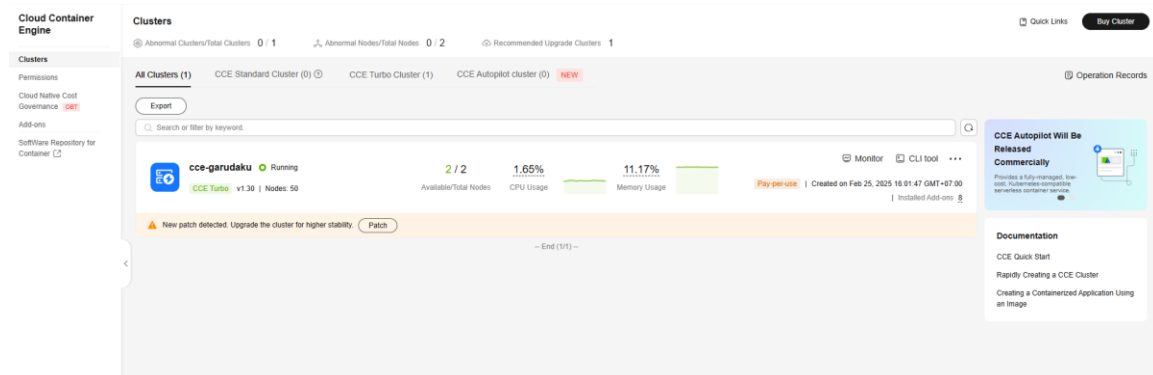
Langkah 3: Push Image ke SWR

```
docker push swr.ap-southeast-3.myhuaweicloud.com/aryadev/websocket-server:latest
```

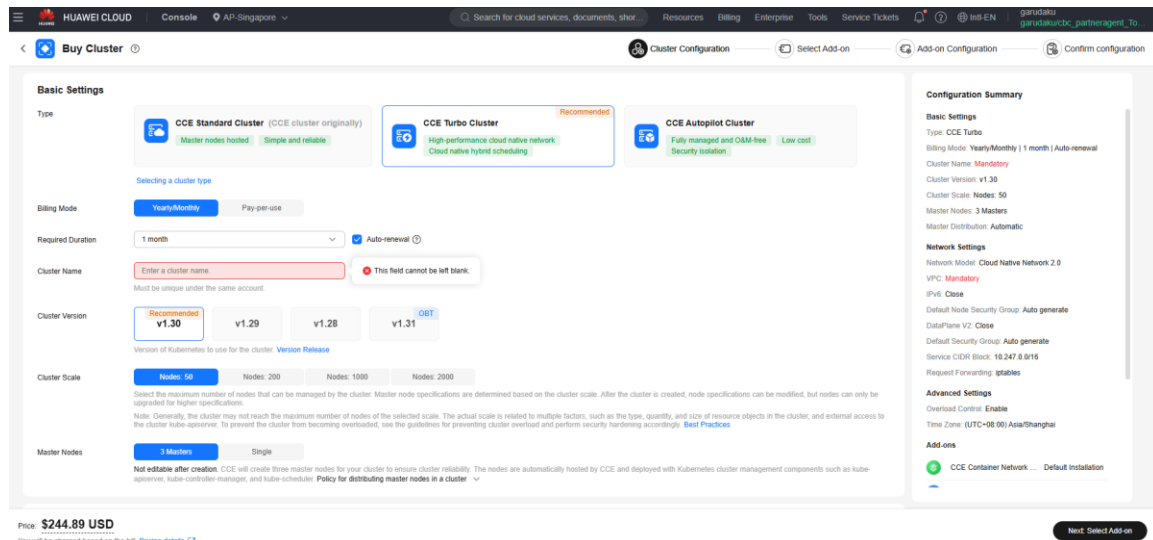
3. Deploy Image ke CCE (Cloud Container Engine)



Langkah 1: Buat Cluster CCE (Jika belum ada)



- Masuk ke CCE > Clusters > Buy Cluster



Basic Setting

- Pilih region: ap-southeast-3 (Singapore).
- Pilih jenis cluster (misal: Standard , Turbo, AutoPilot Cluster).
- Pilih Billing Mode (misal : Yearly/Montly, PPU/PayPerUse).
- Masukkan Nama Cluster.
- Pilih Cluster Scale dan Master Node.

HUAWEI CLOUD Console AP-Singapore

Search for cloud services, documents, short... Resources Billing Enterprise Tools Service Tickets

Buy Cluster Cluster Configuration Select Add-on Add-on Configuration Confirm configuration

Cluster Network

VPC [Create VPC](#)

Not editable after creation. Select a VPC to provide CIDR blocks for the master and worker nodes and pods in your cluster.

Default Node Subnet [Create Subnet](#)

Not editable after creation. Select a subnet in your VPC. Nodes in the cluster will use the IP addresses in this subnet by default. If needed, you can modify the node subnet during the creation of a node or node pool. Master nodes in a cluster are hosted and maintained by CCE. You only need to configure worker nodes.

IPv6 ☐ [Creating an IPv6/IPv4 Dual-Stack Cluster](#)

Default Node Security Group

Two default security groups will be automatically created for your cluster, one for the master nodes and the other for the worker nodes. The security group for the master nodes is named {Cluster name}-cce-control-{Random ID}, and that of the worker nodes is named {Cluster name}-cce-node-{Random ID}. [Learn more about default security group rules.](#)

Container Network

Network Model

☒ **Cloud Native Network 2.0**

Cloud Native Network 2.0 is a next-generation container network model. It deeply integrates the underlying network and offers superb performance.

[Network Model Comparison](#)

Not editable after creation. Model framework used by the container network in the cluster.

DataPlane V2 ☐

DataPlane V2 uses eBPF to enable features like Service ClusterIP, NetworkPolicy, and egress bandwidth. [DataPlane V2 Network Acceleration](#)

Pod Subnet [Create Subnet](#)

CIDR blocks used by pods in the cluster. After a cluster is created, pod CIDR blocks can still be added.

By default, each node in the cluster is pre-bound with 10 ENIs for network acceleration. The IP addresses of these ENIs are allocated from the pod subnet. Properly plan container CIDR blocks and the number of IP addresses. [Best Practices for Configuring NIC Dynamic Preheating](#)

Default Security Group

Price: **\$0.42 USD** Hour

You will be charged based on the bill. [Pricing details](#)

Configuration Summary

Type: CCE Turbo

Billing Mode: Pay-per-use

Cluster Name: **Mandatory**

Cluster Version: v1.30

Cluster Scale: **Nodes: 50**

Master Nodes: **3 Masters**

Master Distribution: **Automatic**

Network Settings

Network Model: **Cloud Native Network 2.0**

VPC: **default_vpc (192.168.0.0/16)**

Default Node Subnet: **Mandatory**

IPv6: **Close**

Default Node Security Group: **Auto generate**

DataPlane V2: **Close**

Pod Subnet: **Mandatory**

Default Security Group: **Auto generate**

Service CIDR Block: **10.247.0.0/16**

Request Forwarding: **iptables**

Advanced Settings

Overload Control: **Enable**

Time Zone: **(UTC+08:00) Asia/Shanghai**

Add-ons

☒ CCE Container Network ... [Default Installation](#)

[Next: Select Add-on](#)

HUAWEI CLOUD Console AP-Singapore

Search for cloud services, documents, short... Resources Billing Enterprise Tools Service Tickets

Buy Cluster Cluster Configuration Select Add-on Add-on Configuration Confirm configuration

Group

Two default security groups will be automatically created for your cluster, one for the master nodes and the other for the worker nodes. The security group for the master nodes is named {Cluster name}-cce-control-{Random ID}, and that of the worker nodes is named {Cluster name}-cce-node-{Random ID}. [Learn more about default security group rules.](#)

Container Network

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By default, each node in the cluster is pre-bound with 10 ENIs for network acceleration. The IP addresses of these ENIs are allocated from the pod subnet. Properly plan container CIDR blocks and the number of IP addresses. [Best Practices for Configuring NIC Dynamic Preheating](#)

Default Security Group

A security group named {Cluster name}-cce-eni-{Random ID} will be created for your cluster and associated with the containers by default. [Learn more about default security group rules.](#)

Service Network

Service CIDR Block

/

Max. Services allowed by this CIDR block: **65,536**

Not editable after creation. Configure an IP address range for ClusterIP Services in your cluster.

Request Forwarding [Comparing iptables and IPVS](#)

Not editable after creation. iptables can be used to balance load between Services and their backend pods stably and reliably.

Use IPVS if there are more than 3,000 Services in your cluster.

Price: **\$0.42 USD** Hour

You will be charged based on the bill. [Pricing details](#)

Configuration Summary

Type: CCE Turbo

Billing Mode: Pay-per-use

Cluster Name: **Mandatory**

Cluster Version: v1.30

Cluster Scale: **Nodes: 50**

Master Nodes: **3 Masters**

Master Distribution: **Automatic**

Network Settings

Network Model: **Cloud Native Network 2.0**

VPC: **default_vpc (192.168.0.0/16)**

Default Node Subnet: **Mandatory**

IPv6: **Close**

Default Node Security Group: **Auto generate**

DataPlane V2: **Close**

Pod Subnet: **Mandatory**

Default Security Group: **Auto generate**

Service CIDR Block: **10.247.0.0/16**

Request Forwarding: **iptables**

Advanced Settings

Overload Control: **Enable**

Time Zone: **(UTC+08:00) Asia/Shanghai**

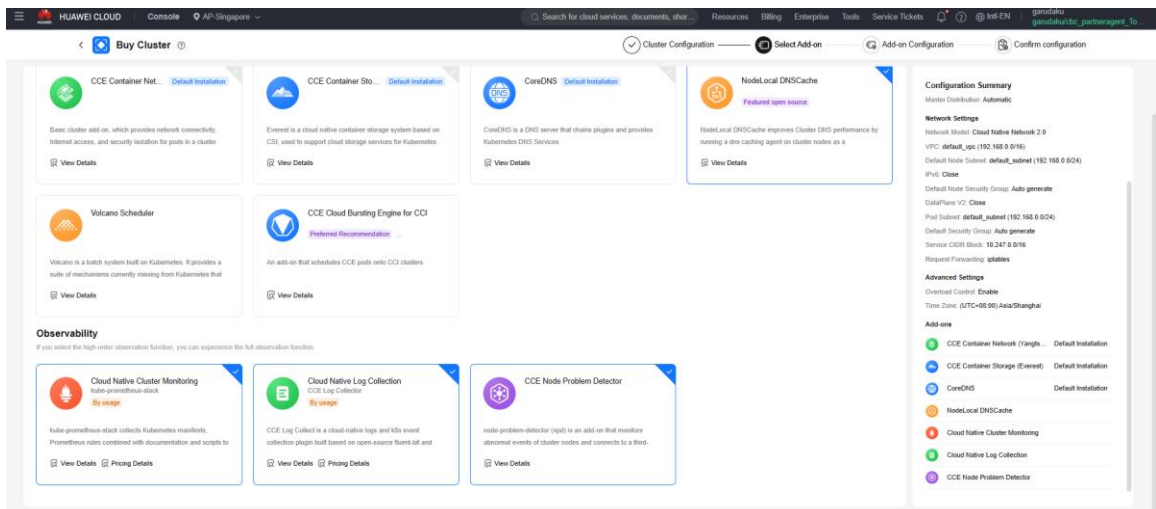
Add-ons

☒ CCE Container Network ... [Default Installation](#)

[Next: Select Add-on](#)

Network Setting

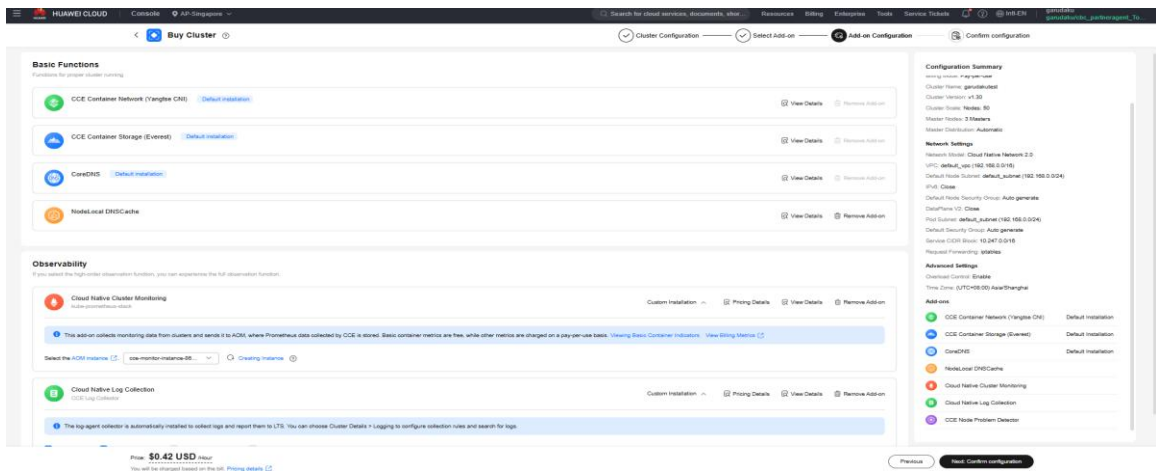
- Pilih VPC yang tersedia (Jika Ada), Kalau tidak ada buat terlebih dahulu untuk VPCnya.
- Pilih Node Subnet yang tersedia (Jika Ada).
- Default Node Security Pilih Auto Generate.
- Pilih Pod Subnet.
- Tentukan Service CIDR Block .
- Next .



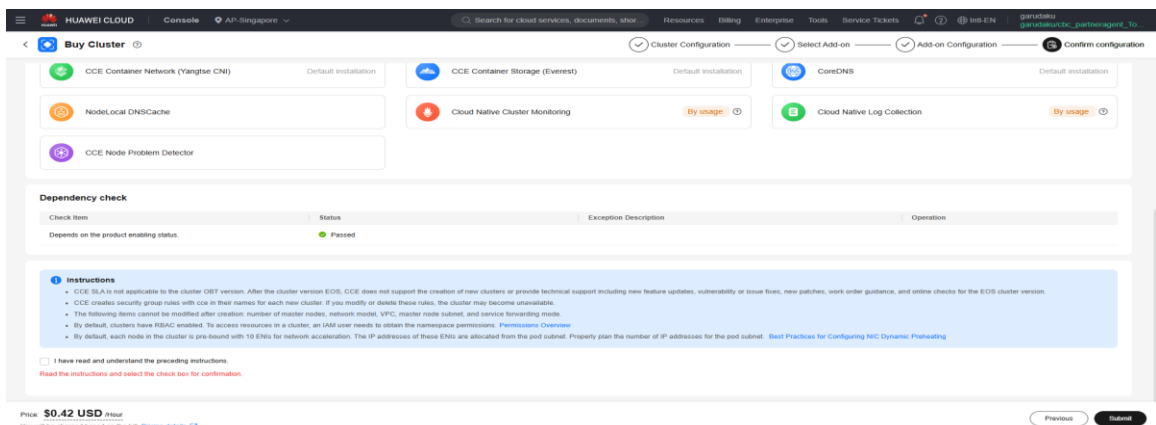
Select Add-on

- Pilih Add-on yang dibutuhkan, Best Practicenya **Default**.

- Next.



- Next.



- Submit.

Langkah 2: Configuration Node Pools dan Nodes

1. Masuk Cluster > Nodes > Create Node Pools

The screenshot displays the Huawei Cloud console interface for managing Kubernetes clusters. The left sidebar shows the navigation menu with 'Nodes' selected. The main content area shows the 'Node Pools' section for the cluster 'cce-garudaku'. Two node pools are listed:

- cce-garudaku-nodepool-messaging** (Normal):
 - Node Type: Elastic Cloud Server (VM)
 - Total number of nodes (actual/expected): 1 / 1
 - CPU Usage/Request: 1.46% / 58.6%
 - Memory Usage/Request: 11.35% / 47.81%
 - Table with 7 columns: Specifications, AZ, Status, Actual/Desired Nodes, Number of yearly/mo..., Pay per Use Nodes, Auto Scaling, Operation.
 - Rows: 5 nodes (AZ1, AZ2, AZ3, AZ5, AZ1) all in 'Normal' status.
- cce-garudaku-nodepool-base** (Normal):
 - Node Type: Elastic Cloud Server (VM)
 - Total number of nodes (actual/expected): 1 / 1
 - CPU Usage/Request: 2.33% / 62.65%
 - Memory Usage/Request: 10.97% / 52.6%
 - Table with 7 columns: Specifications, AZ, Status, Actual/Desired Nodes, Number of yearly/mo..., Pay per Use Nodes, Auto Scaling, Operation.
 - Rows: 2 nodes (AZ1, AZ2) both in 'Normal' status.

- Storage Settings

Configure storage resources for containers and applications on the node

System Disk

General-purpose SSD (AZ3|AZ2|AZ1|AZ5)

50

GB

Expand System Disk Encryption: Not encrypted

System Component Storage

Data Disk

System Disk

Data Disk

General-purpose SSD (AZ3|AZ2|AZ1|AZ5)

100

GB

Quantity

1

Default Data Disk

Used by this container runtime and kubelet. Do not uninstall this disk. Otherwise, the node will become unavailable.
[Choosing a Data Disk Size](#) [How do I allocate data disk space?](#)

Expand Container Engine: Shared disk space | Write Mode: Linear | Data Disk Encryption: Not encrypted

Add

You can add 15 more EVS data disks.

Network Settings

Configure networking resources for node and application communication.

VPC

default_vpc

Node Subnet

Multiple

Single

subnet-master-academy (192.168.16.0/21)Subnet

Available Subnet IP Addresses: 2,034

If the single subnet IP resources associated with your node pool are tight, it is recommended that you configure multiple subnets for the node pool

⚠️ If the default DNS server of the subnet is modified, ensure that the custom DNS server can resolve the OBS service domain name. Otherwise, the node cannot be created.

Node IP

Automatic

Associate Security Group

cce-garuda-cce-node-75qe

Default

Create Security Group

Configure security group rules for worker nodes in your cluster. The rules will take effect on all worker nodes in the cluster. For additional security group configurations, go to the Security Groups page. The security group for master nodes is automatically created by CCE.
[View Default Security Group Rules](#)
Not editable after creation

☐ I have confirmed that the security group rules have been correctly configured for nodes to communicate with each other
- Next Confirm

- Pilih image ECS (Ubuntu/CentOS) dan SSH key.
- Tentukan disk system dan data disk sesuai kebutuhan.

- Klik “Submit” untuk memulai pembuatan cluster.
- Tunggu beberapa menit hingga cluster aktif.

Langkah 3: Buat Deployment di CCE

- Masuk ke Cluster > Namespaces > Create Namespace

The screenshot shows the 'Create Namespace' dialog in the Huawei Cloud console. The 'Name' field is highlighted with a red border and contains the placeholder text 'Enter a namespace name.'. The 'Description (Optional)' field contains the placeholder 'Enter a description.'. The 'Quota Management' toggle is currently turned off. In the background, a table lists existing namespaces:

Namespace	Status	Source	Label
garudaku-message	Running	User	kubernetes.io/metadata.name: project_garudaku-message
monitoring	Running	User	kubernetes.io/metadata.name: name_monitoring
default	Running	Default	kubernetes.io/metadata.name: default; node-local-dns injection: enabled
kube-node-lease	Running	System	kubernetes.io/metadata.name: kube-node-lease
kube-public	Running	System	kubernetes.io/metadata.name: kube-public
kube-system	Running	System	kubernetes.io/metadata.name: kube-system

- Masuk ke Cluster > Workloads > Deployments > Create Deployment

The screenshot shows the 'Create Workload' dialog in the Huawei Cloud console. The 'Workload Type' is set to 'Deployment'. The 'Workload Name' is 'websocket-garudaku'. The 'Namespace' is 'garudaku-message'. The 'Pods' count is 2. The 'Container' is 'runC'. The 'Runtime' is 'Secure runtime'. The 'Time Zone' is set to 'UTC+8:00 (Asia/Singapore)'. The 'Create Workload' button is visible at the bottom right.

- Isi detail Basic Setting:

- Workload Type : Deployment.
- Workload Name: websocket-garudaku.
- Pods: misal 2.

The screenshot shows the 'Container Settings' page in the Huawei Cloud Console. The page is divided into several sections:

- Low-Priority Service:** A toggle switch is turned off. A warning message states: 'Low-priority services cannot be enabled because the Volcano plug-in is not installed in the cluster. [Install Add-on](#)'.
- Time Zone Synchronization:** A toggle switch is turned off. A message states: 'Allows containers to use the same time zone as the node where they run. (This function is realized by the local disks mounted to the containers. Do not modify or delete the local disks.)'
- Container Information:**
 - Container - 1:** The container name is 'container-1'.
 - Basic Info:**
 - Image Name:** A dropdown menu with 'Select Image'.
 - Image Tag:** A dropdown menu with '-Select-'.
 - CPU Quota:** Request: 0.25 cores, Limit: 0.25 cores.
 - Memory Quota:** Request: 512.00 MB, Limit: 512.00 MB.
 - GPU Quota:** A message states: 'This function is unavailable because add-on GPU is not installed. [Install Add-on](#)'.
 - NPU Quota:** A message states: 'This function is unavailable because add-on NPU is not installed. [Install Add-on](#)'.
 - Privileged Container:** A toggle switch is turned off.
 - Image Access Credential:** A dropdown menu with 'default-secret' and a 'Create Secret' button.
 - GPU:** A dropdown menu with 'All'.

- Container Settings:

- Container Name : Isi sesuai kebutuhan.
- Image Name > Klik Select Image > Pilih Image yang sebelumnya di Push ke SWR.
- Image Tag > Pilih Image Tag yang sebelumnya di Push ke SWR.
- CPU dan Memory disesuaikan dengan kebutuhan.
- Untuk settingan berikutnya biarkan Default dan sesuaikan dengan kebutuhan Jika ingin menggunakan PV dan PVC (Container Settings > Data Storage > Add Volume).
- Jika semua sudah disesuaikan dengan kebutuhan, klik create workload.

The screenshot shows the 'Workload List' page in the Huawei Cloud Console. The page displays the following information:

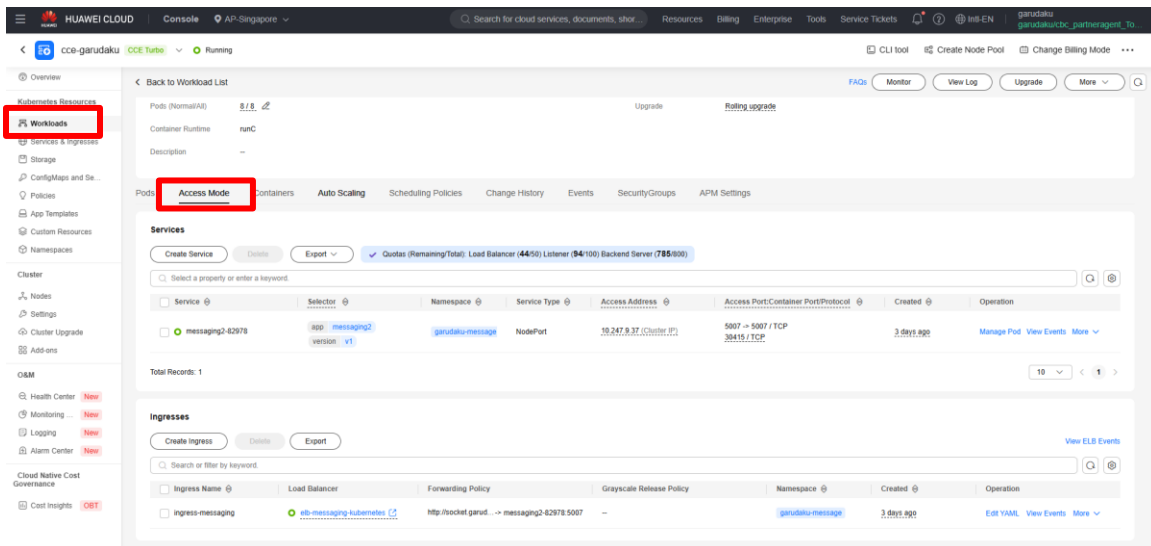
- Workload Name:** socketku
- Status:** Running
- Pods (Normal/All):** 2/2
- Container Runtime:** runc
- Description:** -
- Namespace:** default
- Created:** 12 days ago
- Upgrade:** Rolling upgrade

Below the workload information, there is a table showing the details of the pods:

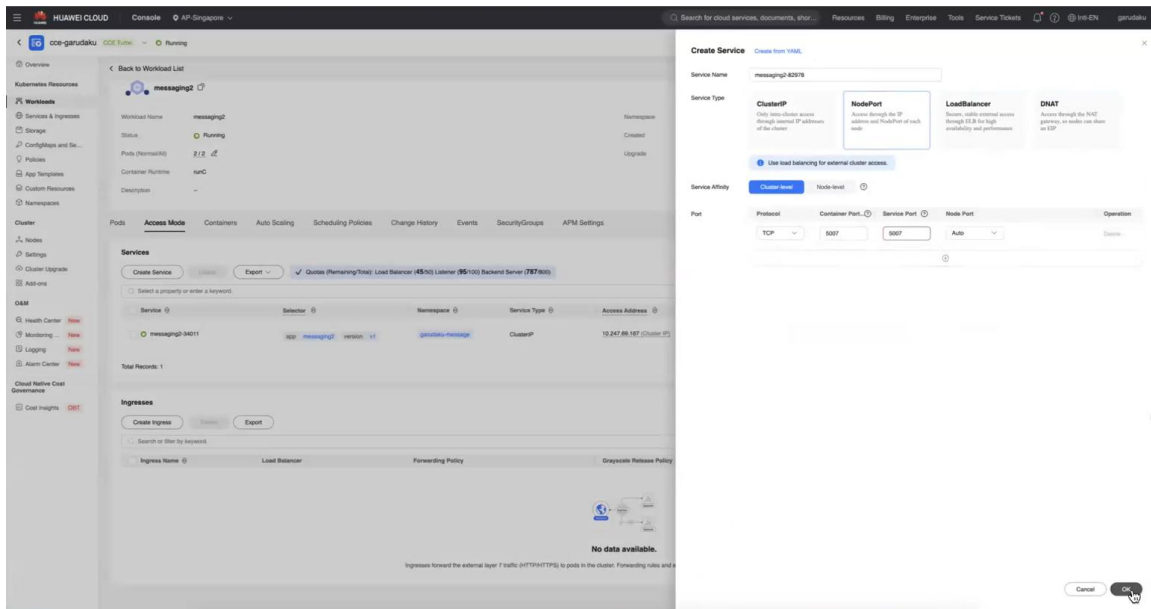
Pod Name	Status	Namespace	Pod IP	Node	Restarts	CPU	Memory	Created	Operation
socketku-68575587c-thuy	Running	default	192.168.38.179	192.168.19.163	0	0.25 Cores 0.5 Cores 0.62%	512 MB 1 GB 8.62%	10 days ago	Monitor More
socketku-68575587c-vpvg	Running	default	192.168.24.123	192.168.19.163	0	0.25 Cores 0.5 Cores 0.62%	512 MB 1 GB 8.57%	10 days ago	Monitor More

Total Records: 2

Langkah 4: Buat Service dan Ingress



- Masuk ke Workloads > Access Mode > Services > Create Service



- Port Mapping:

- Service Port: 80
- ContainerPort: 5007

Catatan Tambahan: Jenis Service Type di CCE Huawei Cloud

1. ClusterIP (default):

- Hanya dapat diakses di dalam cluster.
- Cocok untuk komunikasi antar layanan internal (misalnya: database, backend service).

2. NodePort:

- Mengekspos layanan melalui IP Node dan port tertentu (30000–32767).

- Dapat diakses dari luar cluster dengan cara: `http://<node-ip>:<nodeport>`.
- Tidak scalable otomatis, tergantung IP Node ECS.

3. LoadBalancer (Direkomendasikan):

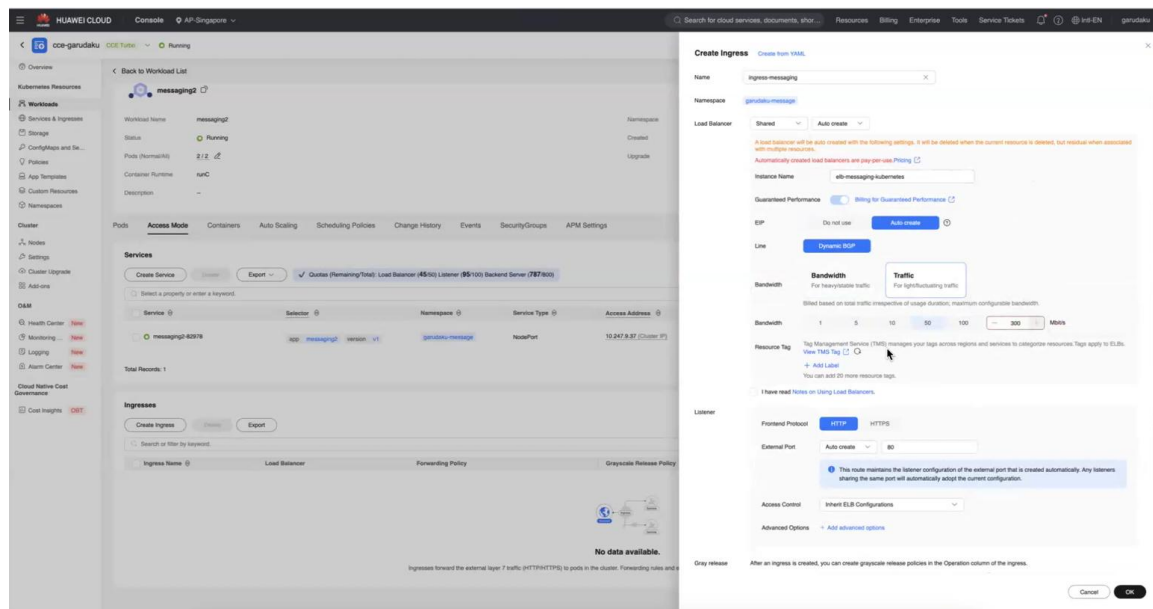
- Membuat Elastic Load Balancer (ELB) secara otomatis.
- Cocok untuk mengakses aplikasi dari luar cluster (publik).
- Akses melalui: `http://<public-ip>:<port>` atau `ws://<public-ip>:80`.
- Mendukung akses wss/https dengan konfigurasi tambahan.

5. Ingress (Opsional):

- Digunakan untuk HTTP/HTTPS routing berbasis path atau domain.
- Memerlukan Ingress Controller seperti NGINX.
- Cocok untuk kebutuhan reverse proxy atau routing domain.

Rekomendasi untuk WebSocket: Gunakan Service Type: LoadBalancer. Jika butuh routing domain atau TLS, tambahkan Ingress Controller.

- Masuk ke Workloads > Access Mode > Ingress > Create Ingress



- Nama Ingress : Disesuaikan
 - Namespace : Pastikan namespace tidak salah
 - Tipe: LoadBalancer (Share/Dedicated – Auto Create)
- Jika Pilih Shared
- Pilih deployment websocket-deployment
 - Instance Name : Disesuaikan kebutuhan.
 - Frontend Subnet : Disesuaikan kebutuhan.

- Backend Subnet : Disesuaikan kebutuhan.

- EIP : Auto create

Jika Pilih ELB Shared

Instance name : Disesuaikan dengan kebutuhan

Eip : Auto Create -> Line : Dynamic BGP -> Bandwidth : Traffic 300Mb ->

The screenshot displays the AWS Management Console configuration for an ELB. The 'Listener' section is active, showing 'Frontend Protocol' set to 'HTTP' and 'External Port' set to 'Auto create' with a value of '80'. A blue informational box states: 'This route maintains the listener configuration of the external port that is created automatically. Any listeners sharing the same port will automatically adopt the current configuration.' Below this, 'Access Control' is set to 'Inherit ELB Configurations'. The 'Forwarding Policy' section shows a table with columns: 'Domain Name', 'Path Match', 'Path', 'Destination Subnet', 'Destination Subnet', 'Set ELB', and 'Operations'. The first row has 'Enter a domain' in the 'Domain Name' field, 'Prefix' in the 'Path Match' dropdown, and 'Enter a path' in the 'Path' field. The 'Set ELB' column has a 'Custom' button, and the 'Operations' column has a 'Delete' button. At the bottom, there is an 'Annotation' section with 'Key' and 'Value' fields, a 'Confirm' button, and 'Cancel' and 'OK' buttons at the bottom right.

Listener ELB :

- Frontend Protocol : HTTP/HTTPS (Untuk HTTPS requirementnya harus menggunakan SSL)

- External Port : Auto Create = 80

- Access Control : Inherit ELB Configuration

The screenshot shows the 'Forwarding Policy' configuration in the AWS Management Console. The 'Domain Name' field contains 'socket.ga'. The 'Path Match' dropdown is open, showing options: 'Prefix' (selected), 'Exact', and 'RegEx'. The 'Path' field contains 'Enter a path'. The 'Destination Subnet' and 'Destination Subnet' fields both have '--Select--' dropdowns. The 'Set ELB' column has a 'Custom' button, and the 'Operations' column has a 'Delete' button. At the bottom, there is an 'Annotation' section with 'Key' and 'Value' fields, a 'Confirm' button, and 'Cancel' and 'OK' buttons at the bottom right.

Fording policy

Domain Name : Disesuaikan dengan kebutuhan

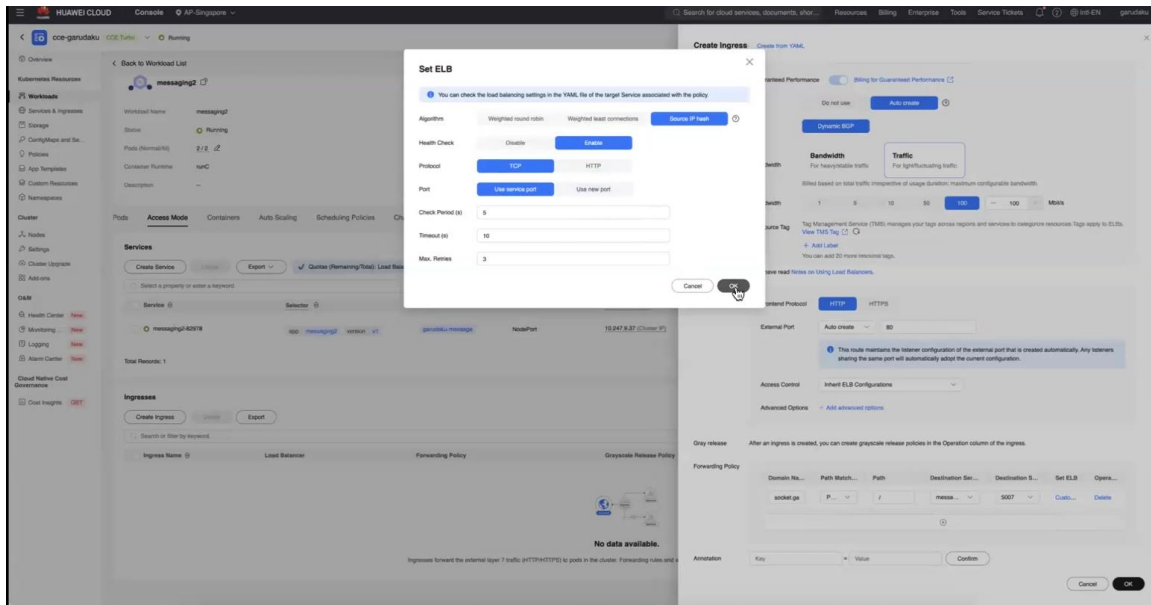
Path Match : Prefix Match

Path : /

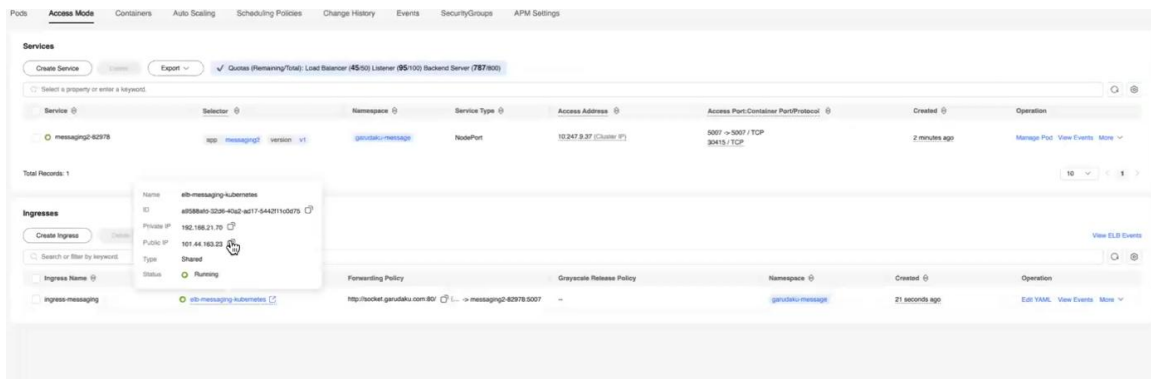
Destination Service : Disesuaikan dengan Service yang sebelumnya telah dibuat

Port : 5007

Set ELB : untuk Web Socket settingannya seperti gambar dibawah ini.



Klik OK. Setelah Service berhasil dibuat, akan muncul public IP address untuk akses WebSocket Anda.



4. Pengujian

- Jika menggunakan domain, maka perlu daftar IP public ELB ke DNS.
- Akses melalui: ws://<public-ip>:80 atau wss:// jika menggunakan HTTPS
- Pastikan aplikasi WebSocket sudah berjalan normal.

